adrianneftali.s@gmail.com
(+52) 951-456-2184
San Diego Area / Ensenada

Adrian Neftali Sanchez

Senior Embedded & Wireless Systems Engineer

linkedin.com/in/adrian-neftali-sanchez
github.com/lutgaru

Professional Summary

Senior Embedded & Wireless Systems Engineer with 4+ years of hands-on experience architecting production-grade firmware, low-latency software, and local-first Edge AI architectures. Specialized in multi-role BLE, low-power optimization, and systems-level programming in C/C++ and Rust. Co-inventor of a patented maritime wireless link integrity system. Proven track record of delivering up to 50% power reduction across 5,000+ deployed commercial units. Eligible for immediate US employment under the TN/T-MEC visa framework.

Technical Skills

Languages & Core:	C, C++, Rust, Python, JavaScript, Shell Scripting
Embedded & RTOS:	Zephyr RTOS, Bare-Metal, Custom RTOS, Bootloaders, Cross-compilers, Linux Kernel Modules
Wireless & Protocols:	Bluetooth Low Energy (Multi-role), Custom Protocol Design, OAD/OTA, CAN (NMEA 2000), SPI, I2C, UART
AI & Edge Computing:	Local Inference, llama.cpp, Vulkan acceleration, Multimodal LLM Integration, Audio Streams
DevOps & Tooling:	Git, CI/CD, GitHub Actions, Docker, Tauri v2, eframe, Renode, Logic Analyzers, Oscilloscopes

Professional Experience

Senior R&D Engineer – Embedded & Wireless Systems — Jun 2022 — Present

Navico Group

- **Co-developed and shipped** production-grade firmware for the Lowrance RECON trolling motor, resulting in **5,000+ commercially deployed units**.
- **Achieved a 50% reduction in power consumption** and **optimized firmware memory footprint by 33%** through targeted bare-metal and software optimizations.
- **Architected a multi-role BLE subsystem** supporting up to 5 simultaneous peripherals and 1 mobile master device using TI BLE stack and Zephyr RTOS.
- **Designed and implemented a custom BLE application-layer protocol** from scratch, standardizing communication for production devices.
- **Co-designed an Over-the-Air (OAD/OTA) firmware update architecture**, ensuring secure and reliable field updates.
- **Led technical code reviews** for over 100 pull requests in the Bluetooth subsystem and authored 80% of official wireless documentation.
- **Collaborated in a distributed international engineering team** across 3 cities and 2 countries, integrating NMEA 2000 CAN-based marine networks.

Engineering Services – Satellite Ground Station Deployment — Mar 2019 — Aug 2019

CICESE

- **Co-deployed and commissioned** the automated ground station infrastructure for the Mexican nanosatellite Painani 1.
- **Developed custom automation software** in Python/C for antenna rotor tracking mechanisms and RF signal decoding via SatNOGS.

Professional Internship – Telecommunications Systems — Aug 2018 — Dec 2018

CICESE

- **Co-inventor of Mexican Patent No. 430014**, featuring an automatic link integrity compensation system under dynamic wave-induced motion.
- **Designed and validated a custom long-range radio protocol** for oceanographic buoy telemetry under real-world maritime field tests.

Featured Open Source Projects

Babilo: High-Performance Local-First Multimodal AI Engine — [GitHub]

Lead Developer — Rust, Tauri v2, Vulkan, C++ (llama.cpp)

- **Architected a low-latency, 100% offline AI voice agent** utilizing an end-to-end multimodal pipeline

$$\text{Voice Input} \rightarrow \text{Multimodal LLM} \rightarrow \text{TTS Output}$$

.
- **Implemented native audio buffer orchestration** and real-time streaming in Rust, completely removing intermediate model friction.
- **Integrated Vulkan-accelerated local inference** via `llama.cpp` to run 9B parameter multimodal models directly on edge hardware.
- **Designed a lightweight cross-platform container** using Tauri v2 and optimized frontend rendering with Vite + Lit Web Components.

SerialGUI-rs: Cross-Platform Graphical Serial Terminal [GitHub]
Lead Developer — Rust, eframe, serialport-rs

- **Developed a lightweight, native GUI tool** for real-world hardware engineering workflows and serial port telemetry monitoring.
- **Leveraged Rust's memory safety guarantees** to implement stable, real-time, cross-platform serial communication handling.

Education

M.Sc. in Electronics and Telecommunications Aug 2019 — Mar 2022
CICESE

- **Thesis:** *Evaluation of encryption overhead in multihop networks for the Internet of Medical Things (IoMT).* Benchmarked 8 NIST Lightweight Cryptography candidates against AES using TI CC2650 hardware and Renode simulation frameworks.

B.Sc. in Electronic Engineering Aug 2014 — Aug 2018
Instituto Tecnológico de Oaxaca